Lexical Grammar

Logan Thomas

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1 Ambiguity

Ambiguity is recognized between some of the regexes listed below. This ambiguity is solved in the lexer by taking the longest match or through regex precedence. The longest match is to pick the regex that matches the most characters (greedy matching), while the precedence resolution is to define a precedence of regexes such that in the case where two regexes match the same span one is preferred.

1.1 Examples

Some examples of ambiguous matches include:

- ”else” matches both $< else >$ and $< ident >$. Solved with precedence.
- ”1.2e3” matches as either $< float\_const >$ followed by an $< ident >$ or as a $< scientific >$. Solved with longest match.
- ”r\”abc\”” matches as either an $< ident >$ followed by a $< string\_literal >$ or as just a $< string\_literal >$. Solved with longest match.

2 Regex

2.1 Substitutions

$$
\begin{align}
L &:= \mathrm{[a-z]} \tag{1}\\
U &:= \mathrm{[A-Z]} \tag{2}\\
D &:= \mathrm{[0-9]} \tag{3}\\
H &:= \mathrm{[a-fA-F0-9]} \tag{4}\\
O &:= \mathrm{[0-7]} \tag{5}\\
B &:= \mathrm{[01]} \tag{6}
\end{align}
$$

2.2 Identifiers, Literals, and Comments

$\_\{0,2\}L(?:L\|U\|D\|\_)*$	$\rightarrow$	IDENT		(7)
$U(?:L\|U\|D)*$	$\rightarrow$	TYPE_IDENT		(8)
$[\backslash+-]?D+$	$\rightarrow$	INTEGER_CONST		(9)
$0[\mathrm{xX}](H\{1,16\})$	$\rightarrow$	NUMERIC_CONST	Hex	(10)
$0[\mathrm{oO}](1O\{21\}\|O\{1,21\})$	$\rightarrow$	NUMERIC_CONST	Octal	(11)
$0[\mathrm{bB}](B\{1,64\})$	$\rightarrow$	NUMERIC_CONST	Binary	(12)
$[\backslash+-]?(?:D+\backslash.D*\|\backslash.D+)$	$\rightarrow$	FLOAT_CONST	Float	(13)
$([\backslash+-]?(?:D+(?:\backslash.D*)?\|\backslash.D+))[eE]([\backslash+-]?D+)$	$\rightarrow$	NUMERIC_CONST	Scientific	(14)
$(\mathrm{r}?)"((?:[\hat{}\ "\backslash\backslash]\|\backslash\backslash")*)"$	$\rightarrow$	STRING_LITERAL		(15)
$\#.*\$$	$\rightarrow$	COMMENT		(16)

2.3 Symbols

:	$\rightarrow$	COLON	(17)
,	$\rightarrow$	COMMA	(18)
{	$\rightarrow$	LEFT_CURLY	(19)
(	$\rightarrow$	LEFT_PAREN	(20)
[	$\rightarrow$	LEFT_SQUARE	(21)
}	$\rightarrow$	RIGHT_CURLY	(22)
)	$\rightarrow$	RIGHT_PAREN	(23)
]	$\rightarrow$	RIGHT_SQUARE	(24)
;	$\rightarrow$	SEMI	(25)

2.4 Keywords

and	→	AND	(26)
bool	→	BOOL	(27)
break	→	BREAK	(28)
class	→	CLASS	(29)
elif	→	ELIF	(30)
else	→	ELSE	(31)
enum	→	ENUM	(32)
exit	→	EXIT	(33)
false	→	FALSE	(34)
FLOAT32_MAX	→	FLOAT32_T_MAX	(35)
FLOAT32_MIN	→	FLOAT32_T_MIN	(36)
float32	→	FLOAT32_T	(37)
FLOAT64_MAX	→	FLOAT64_T_MAX	(38)
FLOAT64_MIN	→	FLOAT64_T_MIN	(39)
float64	→	FLOAT64_T	(40)
NaN	→	FLOAT_NAN	(41)
for	→	FOR	(42)
func	→	FUNCTION	(43)
if	→	IF	(44)
INT16_MAX	→	INT16_MAX	(45)
INT16_MIN	→	INT16_MIN	(46)
int16	→	INT16	(47)
INT32_MAX	→	INT32_MAX	(48)
INT32_MIN	→	INT32_MIN	(49)
int32	→	INT32	(50)
INT64_MAX	→	INT64_MAX	(51)
INT64_MIN	→	INT64_MIN	(52)
int64	→	INT64	(53)
INT8_MAX	→	INT8_MAX	(54)
INT8_MIN	→	INT8_MIN	(55)
int8	→	INT8	(56)
iface	→	INTERFACE	(57)
in	→	IN	(58)
mut	→	MUTABLE	(59)
NINF	→	NEG_INF	(60)
not	→	NOT	(61)
or	→	OR	(62)
INF	→	POS_INF	(63)
private	→	PRIVATE	(64)
public	→	PUBLIC	(65)
return	→	RETURN	(66)
string	→	STRING	(67)

true	$\rightarrow$	TRUE	(68)
UINT16_MAX	$\rightarrow$	UINT16_MAX	(69)
UINT16_MIN	$\rightarrow$	UINT16_MIN	(70)
uint16	$\rightarrow$	UINT16	(71)
UINT32_MAX	$\rightarrow$	UINT32_MAX	(72)
UINT32_MIN	$\rightarrow$	UINT32_MIN	(73)
uint32	$\rightarrow$	UINT32	(74)
UINT64_MAX	$\rightarrow$	UINT64_MAX	(75)
UINT64_MIN	$\rightarrow$	UINT64_MIN	(76)
uint64	$\rightarrow$	UINT64	(77)
UINT8_MAX	$\rightarrow$	UINT8_MAX	(78)
UINT8_MIN	$\rightarrow$	UINT8_MIN	(79)
uint8	$\rightarrow$	UINT8	(80)
while	$\rightarrow$	WHILE	(81)

2.5 Operators

$\&=$	$\rightarrow$	AMP_EQUAL	(82)
$\&$	$\rightarrow$	AMP	(83)
$\hat{}\hat{}=$	$\rightarrow$	CARET_CARET_EQUAL	(84)
$\hat{}\hat{}$	$\rightarrow$	CARET_CARET	(85)
$\hat{}=$	$\rightarrow$	CARET_EQUAL	(86)
$\hat{}$	$\rightarrow$	CARET	(87)
$.$	$\rightarrow$	DOT	(88)
$==$	$\rightarrow$	EQUAL_EQUAL	(89)
$=$	$\rightarrow$	EQUAL	(90)
$!=$	$\rightarrow$	EXCLAIM_EQUAL	(91)
$/=$	$\rightarrow$	FSLASH_EQUAL	(92)
$//=$	$\rightarrow$	FSLASH_FSLASH_EQUAL	(93)
$//$	$\rightarrow$	FSLASH_FSLASH	(94)
$/$	$\rightarrow$	FSLASH	(95)
$<=$	$\rightarrow$	LANGLE_EQUAL	(96)
$<<$	$\rightarrow$	LANGLE_LANGLE	(97)
$<$	$\rightarrow$	LANGLE	(98)
$-=$	$\rightarrow$	MINUS_EQUAL	(99)
$--$	$\rightarrow$	MINUS_MINUS	(100)
$-$	$\rightarrow$	MINUS	(101)
$\%=$	$\rightarrow$	PERCENT_EQUAL	(102)
$\%$	$\rightarrow$	PERCENT	(103)
$\vert =$	$\rightarrow$	PIPE_EQUAL	(104)
$\vert$	$\rightarrow$	PIPE	(105)
$+=$	$\rightarrow$	PLUS_EQUAL	(106)
$++$	$\rightarrow$	PLUS_PLUS	(107)
$+$	$\rightarrow$	PLUS	(108)
$?$	$\rightarrow$	QUESTION	(109)
$>=$	$\rightarrow$	RANGLE_EQUAL	(110)
$>>$	$\rightarrow$	RANGLE_RANGLE	(111)
$>$	$\rightarrow$	RANGLE	(112)
$*=$	$\rightarrow$	STAR_EQUAL	(113)
$*$	$\rightarrow$	STAR	(114)
$\sim=$	$\rightarrow$	TILDE_EQUAL	(115)
$\sim$	$\rightarrow$	TILDE	(116)

3 Backus-Naur Form

$$< ident > \quad ::= \quad < lower > \; < ident? > \tag{117}$$
$$\mid \; "\_" \; < lower > \; < ident? > \tag{118}$$
$$\mid \; "\_" \; "\_" \; < lower > \; < ident? > \tag{119}$$
$$< ident? > \quad ::= \quad < lower > \; < ident? > \tag{120}$$
$$\mid < upper > \; < ident? > \tag{121}$$
$$\mid < digit > \; < ident? > \tag{122}$$
$$\mid \; "\_" \; < ident? > \tag{123}$$
$$\mid \; \varepsilon \tag{124}$$
$$< type\_ident > \quad ::= \quad < upper >< type\_ident? > \tag{125}$$
$$< type\_ident? > \quad ::= \quad < type\_ident > \tag{126}$$
$$\mid < lower > \; < type\_ident? > \tag{127}$$
$$\mid < digit > \; < type\_ident? > \tag{128}$$
$$\mid \; \varepsilon \tag{129}$$
$$< float\_const > \quad ::= \quad < scientific > \tag{130}$$
$$\mid < mantissa > \tag{131}$$
$$\mid < decimal\_const > \tag{132}$$
$$\mid < hex\_const > \tag{133}$$
$$\mid < octal\_const > \tag{134}$$
$$\mid < binary\_const > \tag{135}$$
$$< integer\_const > \quad ::= \quad < decimal\_const > \tag{136}$$
$$\mid < hex\_const > \tag{137}$$
$$\mid < octal\_const > \tag{138}$$
$$\mid < binary\_const > \tag{139}$$
$$< comment > \quad ::= \quad "\#" < comment\_text > \tag{140}$$
$$< comment\_text > \quad ::= \quad < comment\_char >< comment\_text > \tag{141}$$
$$< comment\_char > \quad ::= \quad < lower > \mid < upper \mid < digit > \mid < symbol > \mid < non\_newline > \tag{142}$$
$$< string\_literal > \quad ::= \quad < r? >< quote >< content >< quote > \tag{143}$$
$$< content > \quad ::= \quad < char >< content > \mid \; \varepsilon \tag{144}$$
$$< char > \quad ::= \quad < escaped\_quote > \mid < normal\_symbol > \tag{145}$$
$$< scientific > \quad ::= \quad < mantissa >< exponent > \tag{146}$$
$$\mid < decimal\_const >< exponent > \tag{147}$$
$$< mantissa > \quad ::= \quad < sign? >< digits > \; "." \; < digits? > \tag{148}$$
$$\mid < sign? > \; "." \; < digits > \tag{149}$$
$$< exponent > \quad ::= \quad "e" \; < decimal\_const > \tag{150}$$
$$< decimal\_const > \quad ::= \quad < sign? >< digits > \tag{151}$$
$$< hex\_const > \quad ::= \quad "0" \; "x" < hs > \tag{152}$$
$$\mid \; "0" \; "X" < hs > \tag{153}$$
$$< octal\_const > \quad ::= \quad "0" \; "o" < os > \tag{154}$$
$$\mid \; "0" \; "O" < os > \tag{155}$$
$$< binary\_const > \quad ::= \quad "0" \; "b" < bs > \tag{156}$$
$$\mid \; "0" \; "B" < bs > \tag{157}$$

$$< and > \quad ::= \quad "a""n""d" \tag{158}$$

$$< bool > \quad ::= \quad "b""o""o""l" \tag{159}$$

$$< break > \quad ::= \quad "b""r""e""a""k" \tag{160}$$

$$< class > \quad ::= \quad "c""l""a""s""s" \tag{161}$$

$$< elif > \quad ::= \quad "e""l""i""f" \tag{162}$$

$$< else > \quad ::= \quad "e""l""s""e" \tag{163}$$

$$< enum > \quad ::= \quad "e""n""u""m" \tag{164}$$

$$< exit > \quad ::= \quad "e""x""i""t" \tag{165}$$

$$< false > \quad ::= \quad "f""a""l""s""e" \tag{166}$$

$$< float32 > \quad ::= \quad "f""l""o""a""t""3""2" \tag{167}$$

$$< float64 > \quad ::= \quad "f""l""o""a""t""6""4" \tag{168}$$

$$< for > \quad ::= \quad "f""o""r" \tag{169}$$

$$< func > \quad ::= \quad "f""u""n""c" \tag{170}$$

$$< if > \quad ::= \quad "i""f" \tag{171}$$

$$< iface > \quad ::= \quad "i""f""a""c""e" \tag{172}$$

$$< in > \quad ::= \quad "i""n" \tag{173}$$

$$< int16 > \quad ::= \quad "i""n""t""1""6" \tag{174}$$

$$< int32 > \quad ::= \quad "i""n""t""3""2" \tag{175}$$

$$< int64 > \quad ::= \quad "i""n""t""6""4" \tag{176}$$

$$< int8 > \quad ::= \quad "i""n""t""8" \tag{177}$$

$$< mut > \quad ::= \quad "m""u""t" \tag{178}$$

$$< or > \quad ::= \quad "o""r" \tag{179}$$

$$< private > \quad ::= \quad "p""r""i""v""a""t""e" \tag{180}$$

$$< public > \quad ::= \quad "p""u""b""l""i""c" \tag{181}$$

$$< string > \quad ::= \quad "s""t""r""i""n""g" \tag{182}$$

$$< true > \quad ::= \quad "t""r""u""e" \tag{183}$$

$$< uint16 > \quad ::= \quad "u""i""n""t""1""6" \tag{184}$$

$$< uint32 > \quad ::= \quad "u""i""n""t""3""2" \tag{185}$$

$$< uint64 > \quad ::= \quad "u""i""n""t""6""4" \tag{186}$$

$$< uint8 > \quad ::= \quad "u""i""n""t""8" \tag{187}$$

$$< while > \quad ::= \quad "w""h""i""l""e" \tag{188}$$

$< exponent\_assign >$	$::=$	”^” ”^” ”$=$”	(189)
$< exponent >$	$::=$	”^” ”^”	(190)
$< xor\_assign >$	$::=$	”^” ”$=$”	(191)
$< xor >$	$::=$	”^”	(192)
$< mod\_assign >$	$::=$	”%” ”$=$”	(193)
$< mod >$	$::=$	”%”	(194)
$< mul\_assign >$	$::=$	”$*$” ”$=$”	(195)
$< mul >$	$::=$	”$*$”	(196)
$< integer\_div\_assign >$	$::=$	”/” ”/” ”$=$”	(197)
$< integer\_div >$	$::=$	”/” ”/”	(198)
$< div\_assign >$	$::=$	”/” ”$=$”	(199)
$< div >$	$::=$	”/”	(200)
$< inc >$	$::=$	”$+$” ”$+$”	(201)
$< add\_assign >$	$::=$	”$+$” ”$=$”	(202)
$< add >$	$::=$	”$+$”	(203)
$< dec >$	$::=$	”$-$” ”$-$”	(204)
$< sub\_assign >$	$::=$	”$-$” ”$=$”	(205)
$< sub >$	$::=$	”$-$”	(206)
$< shift\_left >$	$::=$	”$<$” ”$<$”	(207)
$< le\_op >$	$::=$	”$<$” ”$=$”	(208)
$< lt\_op >$	$::=$	”$<$”	(209)
$< shift\_right >$	$::=$	”$>$” ”$>$”	(210)
$< ge\_op >$	$::=$	”$>$” ”$=$”	(211)
$< gt\_op >$	$::=$	”$>$”	(212)
$< ne\_op >$	$::=$	”!” ”$=$”	(213)
$< eq\_op >$	$::=$	”$=$” ”$=$”	(214)
$< assign >$	$::=$	”$=$”	(215)
$< and\_assign >$	$::=$	”&” ”$=$”	(216)
$< and >$	$::=$	”&”	(217)
$< or\_assign >$	$::=$	”$\vert$” ”$=$”	(218)
$< or >$	$::=$	”$\vert$”	(219)
$< ternary >$	$::=$	”?”	(220)
$< dot >$	$::=$	”.”	(221)

$$
\begin{align}
<digits?> \quad &::= \quad <digits> \mid \varepsilon \tag{222}\\
<digits> \quad &::= \quad <digit> \mid <digit><digits> \tag{223}\\
<sign?> \quad &::= \quad <sign> \mid \varepsilon \tag{224}\\
<sign> \quad &::= \quad ''+'' \mid ''-'' \tag{225}\\
<lower> \quad &::= \quad ''a''|''b''|''c''|''d''|''e''|''f''|''g''|''h''|''i''|''j''|''k''|''l''|''m''| \notag\\
&\quad\quad ''n''|''o''|''p''|''q''|''r''|''s''|''t''|''u''|''v''|''w''|''x''|''y''|''z'' \tag{226}\\
<upper> \quad &::= \quad ''A''|''B''|''C''|''D''|''E''|''F''|''G''|''H''|''I''|''J''|''K''|''L''|''M''| \notag\\
&\quad\quad ''N''|''O''|''P''|''Q''|''R''|''S''|''T''|''U''|''V''|''W''|''X''|''Y''|''Z'' \tag{227}\\
<digit> \quad &::= \quad ''0''|''1''|''2''|''3''|''4''|''5''|''6''|''7''|''8''|''9'' \tag{228}\\
<hex> \quad &::= \quad ''0''|''1''|''2''|''3''|''4''|''5''|''6''|''7''|''8''|''9''| \notag\\
&\quad\quad ''a''|''b''|''c''|''d''|''e''|''f''|''A''|''B''|''C''|''D''|''E''|''F'' \tag{229}\\
<octal> \quad &::= \quad ''0''|''1''|''2''|''3''|''4''|''5''|''6''|''7'' \tag{230}\\
<binary> \quad &::= \quad ''0''|''1'' \tag{231}\\
<symbols> \quad &::= \quad <normal\_symbol> \mid <quote> \mid <backslash> \tag{232}\\
<normal\_symbol> \quad &::= \quad ''!''|''\#''|''\$''|''\%''|''\&''|'''''|''(''|'')''|''*''|''+''|'',''|''-''|''.''| \notag\\
&\quad\quad ''/''|'':''|'';''|''<''|''=''|''>''|''?''|''@''|''[''|'']''|''\hat{}''|''\_''|''`''| \notag\\
&\quad\quad ''\{''|''|''|''\}''|''\sim'' \tag{233}\\
<whitespace> \quad &::= \quad <non\_newline> \mid <newline> \tag{234}\\
<non\_newline> \quad &::= \quad ''\ ''|''\backslash t''|''\backslash r'' \tag{235}\\
<newline> \quad &::= \quad ''\backslash n'' \tag{236}\\
<r?> \quad &::= \quad ''r'' \mid \varepsilon \tag{237}\\
<escaped\_quote> \quad &::= \quad <backslash><quote> \tag{238}\\
<backslash> \quad &::= \quad ''\backslash'' \tag{239}\\
<quote> \quad &::= \quad '''''' \tag{240}\\
& \tag{241}
\end{align}
$$

$$\begin{aligned}
< hs > \quad & ::= \quad < hex > && (242)\\
& \quad\ | < hex >< hs1 > && (243)\\
< hs1 > \quad & ::= \quad < hex > && (244)\\
& \quad\ | < hex >< hs2 > && (245)\\
< hs2 > \quad & ::= \quad < hex > && (246)\\
& \quad\ | < hex >< hs3 > && (247)\\
< hs3 > \quad & ::= \quad < hex > && (248)\\
& \quad\ | < hex >< hs4 > && (249)\\
< hs4 > \quad & ::= \quad < hex > && (250)\\
& \quad\ | < hex >< hs5 > && (251)\\
< hs5 > \quad & ::= \quad < hex > && (252)\\
& \quad\ | < hex >< hs6 > && (253)\\
< hs6 > \quad & ::= \quad < hex > && (254)\\
& \quad\ | < hex >< hs7 > && (255)\\
< hs7 > \quad & ::= \quad < hex > && (256)\\
& \quad\ | < hex >< hs8 > && (257)\\
< hs8 > \quad & ::= \quad < hex > && (258)\\
& \quad\ | < hex >< hs9 > && (259)\\
< hs9 > \quad & ::= \quad < hex > && (260)\\
& \quad\ | < hex >< hs10 > && (261)\\
< hs10 > \quad & ::= \quad < hex > && (262)\\
& \quad\ | < hex >< hs11 > && (263)\\
< hs11 > \quad & ::= \quad < hex > && (264)\\
& \quad\ | < hex >< hs12 > && (265)\\
< hs12 > \quad & ::= \quad < hex > && (266)\\
& \quad\ | < hex >< hs13 > && (267)\\
< hs13 > \quad & ::= \quad < hex > && (268)\\
& \quad\ | < hex >< hs14 > && (269)\\
< hs14 > \quad & ::= \quad < hex > && (270)\\
& \quad\ | < hex >< hs15 > && (271)\\
< hs15 > \quad & ::= \quad < hex > && (272)
\end{aligned}$$

$$< os > \quad ::= \quad < octal > \tag{273}$$
$$| < octal >< os1 > \tag{274}$$
$$< os1 > \quad ::= \quad < octal > \tag{275}$$
$$| < octal >< os2 > \tag{276}$$
$$< os2 > \quad ::= \quad < octal > \tag{277}$$
$$| < octal >< os3 > \tag{278}$$
$$< os3 > \quad ::= \quad < octal > \tag{279}$$
$$| < octal >< os4 > \tag{280}$$
$$< os4 > \quad ::= \quad < octal > \tag{281}$$
$$| < octal >< os5 > \tag{282}$$
$$< os5 > \quad ::= \quad < octal > \tag{283}$$
$$| < octal >< os6 > \tag{284}$$
$$< os6 > \quad ::= \quad < octal > \tag{285}$$
$$| < octal >< os7 > \tag{286}$$
$$< os7 > \quad ::= \quad < octal > \tag{287}$$
$$| < octal >< os8 > \tag{288}$$
$$< os8 > \quad ::= \quad < octal > \tag{289}$$
$$| < octal >< os9 > \tag{290}$$
$$< os9 > \quad ::= \quad < octal > \tag{291}$$
$$| < octal >< os10 > \tag{292}$$
$$< os10 > \quad ::= \quad < octal > \tag{293}$$
$$| < octal >< os11 > \tag{294}$$
$$< os11 > \quad ::= \quad < octal > \tag{295}$$
$$| < octal >< os12 > \tag{296}$$
$$< os12 > \quad ::= \quad < octal > \tag{297}$$
$$| < octal >< os13 > \tag{298}$$
$$< os13 > \quad ::= \quad < octal > \tag{299}$$
$$| < octal >< os14 > \tag{300}$$
$$< os14 > \quad ::= \quad < octal > \tag{301}$$
$$| < octal >< os15 > \tag{302}$$
$$< os15 > \quad ::= \quad < octal > \tag{303}$$
$$| < octal >< os16 > \tag{304}$$
$$< os16 > \quad ::= \quad < octal > \tag{305}$$
$$| < octal >< os17 > \tag{306}$$
$$< os17 > \quad ::= \quad < octal > \tag{307}$$
$$| < octal >< os18 > \tag{308}$$
$$< os18 > \quad ::= \quad < octal > \tag{309}$$
$$| < octal >< os19 > \tag{310}$$
$$< os19 > \quad ::= \quad < octal > \tag{311}$$
$$| < octal >< os20 > \tag{312}$$
$$< os20 > \quad ::= \quad < octal > \tag{313}$$
$$| < octal >< os21 > \tag{314}$$
$$< os21 > \quad ::= \quad < octal > \tag{315}$$

$$
\begin{aligned}
< bs > \quad &::= \quad < binary > && (316)\\
& \quad\quad | < binary >< bs1 > && (317)\\
< bs1 > \quad &::= \quad < binary > && (318)\\
& \quad\quad | < binary >< bs2 > && (319)\\
< bs2 > \quad &::= \quad < binary > && (320)\\
& \quad\quad | < binary >< bs3 > && (321)\\
< bs3 > \quad &::= \quad < binary > && (322)\\
& \quad\quad | < binary >< bs4 > && (323)\\
< bs4 > \quad &::= \quad < binary > && (324)\\
& \quad\quad | < binary >< bs5 > && (325)\\
< bs5 > \quad &::= \quad < binary > && (326)\\
& \quad\quad | < binary >< bs6 > && (327)\\
< bs6 > \quad &::= \quad < binary > && (328)\\
& \quad\quad | < binary >< bs7 > && (329)
\end{aligned}
$$

$$
\begin{aligned}
< bs7 > \quad &::= \quad < binary > && (330)\\
&\quad | < binary >< bs8 > && (331)\\
< bs8 > \quad &::= \quad < binary > && (332)\\
&\quad | < binary >< bs9 > && (333)\\
< bs9 > \quad &::= \quad < binary > && (334)\\
&\quad | < binary >< bs10 > && (335)\\
< bs10 > \quad &::= \quad < binary > && (336)\\
&\quad | < binary >< bs11 > && (337)\\
< bs11 > \quad &::= \quad < binary > && (338)\\
&\quad | < binary >< bs12 > && (339)\\
< bs12 > \quad &::= \quad < binary > && (340)\\
&\quad | < binary >< bs13 > && (341)\\
< bs13 > \quad &::= \quad < binary > && (342)\\
&\quad | < binary >< bs14 > && (343)\\
< bs14 > \quad &::= \quad < binary > && (344)\\
&\quad | < binary >< bs15 > && (345)\\
< bs15 > \quad &::= \quad < binary > && (346)\\
&\quad | < binary >< bs16 > && (347)\\
< bs16 > \quad &::= \quad < binary > && (348)\\
&\quad | < binary >< bs17 > && (349)\\
< bs17 > \quad &::= \quad < binary > && (350)\\
&\quad | < binary >< bs18 > && (351)\\
< bs18 > \quad &::= \quad < binary > && (352)\\
&\quad | < binary >< bs19 > && (353)\\
< bs19 > \quad &::= \quad < binary > && (354)\\
&\quad | < binary >< bs20 > && (355)\\
< bs20 > \quad &::= \quad < binary > && (356)\\
&\quad | < binary >< bs21 > && (357)\\
< bs21 > \quad &::= \quad < binary > && (358)\\
&\quad | < binary >< bs22 > && (359)\\
< bs22 > \quad &::= \quad < binary > && (360)\\
&\quad | < binary >< bs23 > && (361)\\
< bs23 > \quad &::= \quad < binary > && (362)\\
&\quad | < binary >< bs24 > && (363)\\
< bs24 > \quad &::= \quad < binary > && (364)\\
&\quad | < binary >< bs25 > && (365)\\
< bs25 > \quad &::= \quad < binary > && (366)\\
&\quad | < binary >< bs26 > && (367)\\
< bs26 > \quad &::= \quad < binary > && (368)\\
&\quad | < binary >< bs27 > && (369)
\end{aligned}
$$

$$
\begin{aligned}
< bs27 > \quad &::= \quad < binary > && (370)\\
&\quad | < binary >< bs28 > && (371)\\
< bs28 > \quad &::= \quad < binary > && (372)\\
&\quad | < binary >< bs29 > && (373)\\
< bs29 > \quad &::= \quad < binary > && (374)\\
&\quad | < binary >< bs30 > && (375)\\
< bs30 > \quad &::= \quad < binary > && (376)\\
&\quad | < binary >< bs31 > && (377)\\
< bs31 > \quad &::= \quad < binary > && (378)\\
&\quad | < binary >< bs32 > && (379)\\
< bs32 > \quad &::= \quad < binary > && (380)\\
&\quad | < binary >< bs33 > && (381)\\
< bs33 > \quad &::= \quad < binary > && (382)\\
&\quad | < binary >< bs34 > && (383)\\
< bs34 > \quad &::= \quad < binary > && (384)\\
&\quad | < binary >< bs35 > && (385)\\
< bs35 > \quad &::= \quad < binary > && (386)\\
&\quad | < binary >< bs36 > && (387)\\
< bs36 > \quad &::= \quad < binary > && (388)\\
&\quad | < binary >< bs37 > && (389)\\
< bs37 > \quad &::= \quad < binary > && (390)\\
&\quad | < binary >< bs38 > && (391)\\
< bs38 > \quad &::= \quad < binary > && (392)\\
&\quad | < binary >< bs39 > && (393)\\
< bs39 > \quad &::= \quad < binary > && (394)\\
&\quad | < binary >< bs40 > && (395)\\
< bs40 > \quad &::= \quad < binary > && (396)\\
&\quad | < binary >< bs41 > && (397)\\
< bs41 > \quad &::= \quad < binary > && (398)\\
&\quad | < binary >< bs42 > && (399)\\
< bs42 > \quad &::= \quad < binary > && (400)\\
&\quad | < binary >< bs43 > && (401)\\
< bs43 > \quad &::= \quad < binary > && (402)\\
&\quad | < binary >< bs44 > && (403)\\
< bs44 > \quad &::= \quad < binary > && (404)\\
&\quad | < binary >< bs45 > && (405)\\
< bs45 > \quad &::= \quad < binary > && (406)\\
&\quad | < binary >< bs46 > && (407)\\
< bs46 > \quad &::= \quad < binary > && (408)\\
&\quad | < binary >< bs47 > && (409)
\end{aligned}
$$

$$
\begin{aligned}
< bs47 > \quad & ::= \quad < binary > && (410)\\
& \quad | < binary >< bs48 > && (411)\\
< bs48 > \quad & ::= \quad < binary > && (412)\\
& \quad | < binary >< bs49 > && (413)\\
< bs49 > \quad & ::= \quad < binary > && (414)\\
& \quad | < binary >< bs50 > && (415)\\
< bs50 > \quad & ::= \quad < binary > && (416)\\
& \quad | < binary >< bs51 > && (417)\\
< bs51 > \quad & ::= \quad < binary > && (418)\\
& \quad | < binary >< bs52 > && (419)\\
< bs52 > \quad & ::= \quad < binary > && (420)\\
& \quad | < binary >< bs53 > && (421)\\
< bs53 > \quad & ::= \quad < binary > && (422)\\
& \quad | < binary >< bs54 > && (423)\\
< bs54 > \quad & ::= \quad < binary > && (424)\\
& \quad | < binary >< bs55 > && (425)\\
< bs55 > \quad & ::= \quad < binary > && (426)\\
& \quad | < binary >< bs56 > && (427)\\
< bs56 > \quad & ::= \quad < binary > && (428)\\
& \quad | < binary >< bs57 > && (429)\\
< bs57 > \quad & ::= \quad < binary > && (430)\\
& \quad | < binary >< bs58 > && (431)\\
< bs58 > \quad & ::= \quad < binary > && (432)\\
& \quad | < binary >< bs59 > && (433)\\
< bs59 > \quad & ::= \quad < binary > && (434)\\
& \quad | < binary >< bs60 > && (435)\\
< bs60 > \quad & ::= \quad < binary > && (436)\\
& \quad | < binary >< bs61 > && (437)\\
< bs61 > \quad & ::= \quad < binary > && (438)\\
& \quad | < binary >< bs62 > && (439)\\
< bs62 > \quad & ::= \quad < binary > && (440)\\
& \quad | < binary >< bs63 > && (441)\\
< bs63 > \quad & ::= \quad < binary > && (442)\\
& && (443)
\end{aligned}
$$